The canonical function from identifications of types to equivalences

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled thm:idtoeqv (source lines 1715-1721) with its numbered equation defined inside the statement; the proof span 1722-1732 is the author's proof environment, retained line for line.

Editorial note. The proof is complete in place: transport along the universe family, path induction to reflexivity, and the identity-function equivalence.

Editorial note. Revision history: r001 assembly was refused because the example ref was flagged load-bearing and r002 because its kind was outside the policy vocabulary; r003 restates it as a reviewed navigation pointer with a policy-vocabulary kind (the source environment is the author's example, stated in the reason), without changing the retained spans.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the canonical function from identifications of types to equivalences. The univalence axiom stated immediately after is a separate author statement and is not part of the counted claim.

Allowed standard prerequisites (editorial).

- A universe type and the identity type of the universe between two of its elements.
- The identity function on the universe regarded as a type family indexed by the universe, and transport along a path in a type family.
- Path induction with its conversion rule.
- The example that the identity function of a type is an equivalence, with its quasi-inverse structure.

Source reference. Original section 4.8: sources/hott-basics/book-labelnumbers-first-edition.tex, source lines 1687–1687. Parenthetical navigation pointer inside the author proof (total space of the universe family); nothing in the counted proof depends on that section.

Source reference. Original lemma 2.4.7: sources/hott-basics/pinned-original-source.tex, source lines 1100–1100. Author example asserted and proved in place at source lines 1106-1112 (book 2.4.7); cited by the retained proof in its base case. Navigation pointer only: the counted proof is retained verbatim and

no part of it is replaced by the reference; the cited example is itself proved in the source.

Lemma 0.0.1. *For types $A, B : \mathcal{U}$, there is a certain function,*

$$\mathsf{idtoeqv} : (A =_{\mathcal{U}} B) \to (A \simeq B), \tag{0.0.2}$$

defined in the proof.

Proof. We could construct this directly by induction on equality, but the following description is more convenient. Note that the identity function $\mathsf{id}_{\mathcal{U}} : \mathcal{U} \to \mathcal{U}$ may be regarded as a type family indexed by the universe $\mathcal{U}$; it assigns to each type $X : \mathcal{U}$ the type X itself. (When regarded as a fibration, its total space is the type $\sum_{(A:\mathcal{U})} A$ of "pointed types"; see also original section 4.8.) Thus, given a path $p : A =_{\mathcal{U}} B$, we have a transport function $p_* : A \to B$. We claim that p_* is an equivalence. But by induction, it suffices to assume that p is refl_A, in which case $p_* \equiv \mathsf{id}_A$, which is an equivalence by original lemma 2.4.7. Thus, we can define $\mathsf{idtoeqv}(p)$ to be p_* (together with the above proof that it is an equivalence). □